MAT3007 Optimization: Tutorial 6

Duality Theory

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Why Learning Duality?

- The duality theory motivates new LP algorithms (e.g. Interior point method)
- The LP's dual problem helps solving its primal problem (e.g. weak duality & strong duality. To be explained in lecture later).
- The dual problem offers another interpretation of the original problem and carries much useful information for the primal problem.

Different Duality Forms 1

The primal problem.

$$\min_x \ c^T x$$
$$\text{subject to } Ax \geq b$$

Can be written as

$$\min_x \ c^T x + \max_{y \geq 0} y^T (b - Ax)$$

Exchange the order of max and min

$$\max_{y \geq 0} \ b^T y + \min_x x^T (c - A^T y)$$

Finally get the dual problem

$$\max_y \ b^T y$$
$$\text{subject to } A^T y = c$$
$$y \geq 0$$

Different Duality Forms 2

The primal problem.

$$\min_x \ c^T x$$
$$\text{subject to } Ax \leq b \quad x \geq 0$$

Can be written as

$$\min_{x \geq 0} \ c^T x + \max_{y \leq 0} y^T (b - Ax)$$

Exchange the order of max and min

$$\max_{y \leq 0} \ b^T y + \min_{x \geq 0} x^T (c - A^T y)$$

Finally get the dual problem

$$\max_y \ b^T y$$
$$\text{subject to } A^T y \leq c$$
$$y \leq 0$$

General Rule

	Primal			*Dual*	
minimize	$c^T x$		maximize	$b^T y$	
subject to	$a_i^T x \geq b_i,$	$i \in M_1$	subject to	$y_i \geq 0,$	$i \in M_1$
	$a_i^T x \leq b_i,$	$i \in M_2$		$y_i \leq 0,$	$i \in M_2$
	$a_i^T x = b_i,$	$i \in M_3$		y_i *free*,	$i \in M_3$
	$x_j \geq 0,$	$j \in N_1$		$A_j^T y \leq c_j,$	$j \in N_1$
	$x_j \leq 0,$	$j \in N_2$		$A_j^T y \geq c_j,$	$j \in N_2$
	x_j *free*,	$j \in N_3$		$A_j^T y = c_j,$	$j \in N_3$

Table: General rule for formulating a primal-dual pair of LP.

Here a_i is the i_{th} row of A, and A_i is the i_{th} column of A. Each primal constraint corresponds to a dual variable. Each primal variable corresponds to a dual constraint. Equality constraints always correspond to free variables.

Exercise 1

Transform the following to its dual problem.

$$
\begin{aligned}
\text{maximize}_x \quad & x_1 + x_2 + x_3 \\
\text{subject to} \quad & x_2 + 2x_3 \leq 1 \\
& x_1 + 2x_2 + x_3 \leq 1 \\
& 2x_1 \leq 1 \\
& x_1,\ x_2,\ x_3 \geq 0
\end{aligned}
$$

Exercise 1: Solution

$$
\begin{aligned}
\text{minimize}_y \quad & y_1 + y_2 + y_3 \\
\text{subject to} \quad & y_2 + 2y_3 \geq 1 \\
& y_1 + 2y_2 \geq 1 \\
& 2y_1 + y_2 \geq 1 \\
& y_1,\ y_2,\ y_3 \geq 0
\end{aligned}
$$

Exercise 2

Transform the following to its dual problem.

$$
\begin{aligned}
\text{minimize}_x \quad & -5x_1 - 6x_2 - 4x_3 \\
\text{subject to} \quad & x_1 - 2x_2 \geq -1 \\
& -3x_1 + x_2 \leq -2 \\
& -3x_2 - x_3 = -3 \\
& x_1 \ \textit{free}, \ x_2 \geq 0, \ x_3 \leq 0
\end{aligned}
$$

Exercise 2: Solution

$$
\begin{aligned}
\text{maximize}_x \quad & -y_1 - 2y_2 - 3y_3 \\
\text{subject to} \quad & y_1 - 3y_2 = -5 \\
& -2y_1 + y_2 - 3y_3 \leq -6 \\
& -y_3 \geq -4 \\
& y_1 \geq 0, \ y_2 \leq 0, \ y_3 \ \textit{free}
\end{aligned}
$$

Invariance of Transformation

Theorem (Duality under Transformations)

If we transform a linear program to an equivalent one (e.g., by replacing free variables, adding slack variables, etc.), then the dual of the two problems will be equivalent (i.e.,they are either both infeasible, or they have the same optimal cost.).

Theorem (Dual^2)

If we transform the primal to its dual and transform the resulting dual to its dual, then we will obtain a problem equivalent to the primal problem, i.e., the dual of dual is the primal.

Exercise 3

Consider the problem

$$\begin{aligned} \min \quad & c^T x \\ s.t. \quad & Ax \geq b \end{aligned}$$

1. Derive the dual problem of
   - the original problem.
   - the standard form of the original problem, i.e. adding slack variables.
   - the original problem with x replaced by x^+ and x^-.

2. Show that these three dual problems are equal.

Exercise 3: Solution

$$(P1) \quad \begin{aligned} \min \quad & c^T x \\ s.t. \quad & Ax \geq b \end{aligned}$$

$$(D1) \quad \begin{aligned} \max \quad & b^T y \\ s.t. \quad & A^T y = c \\ & y \geq 0 \end{aligned}$$

$$(P2) \quad \begin{aligned} \min \quad & c^T x + 0^T s \\ s.t. \quad & Ax - s = b \\ & s \geq 0 \end{aligned}$$

$$(D2) \quad \begin{aligned} \max \quad & b^T y \\ s.t. \quad & A^T y = c \\ & -I^T y \leq 0 \end{aligned}$$

$$(P3) \quad \begin{aligned} \min \quad & c^T x^+ - c^T x^- \\ s.t. \quad & Ax^+ - Ax^- \geq b \\ & x^+ \geq 0,\ x^- \geq 0 \end{aligned}$$

$$(D3) \quad \begin{aligned} \max \quad & b^T y \\ s.t. \quad & A^T y \leq c \\ & -A^T y \leq -c \\ & y \geq 0 \end{aligned}$$

It's easy to see that three dual problems are equal.

Exercise 4

The optimal solution of LP primal problem is x*=(0,-0.5,-4.5)^T

$$\begin{aligned} \max_x \quad & -2x_1 - 2x_2 + x_3 \\ \text{subject to} \quad & x_1 + x_2 - x_3 \geq 4 \\ & x_1 + Kx_2 - x_3 \leq 6 \\ & x_1 \geq 0,\ x_2 \leq 0,\ x_3 \leq 0 \end{aligned}$$

1.What is the value of K?
2.Get the LP dual problem.

Exercise 4 Solution

Since the optimal solution of LP primal problem is $x^* = (0, -0.5, -4.5)^T$, the optimal objective value is -2*(0)-2*(-0.5)+(-4.5)=-3.5
The LP dual formulation is

$$\begin{aligned} \text{Min}_y \ & 4y_1 + 6y_2 \\ \text{subject to } & y_1 + y_2 \geq -2 \\ & y_1 + Ky_2 \leq -2 \\ & -y_1 - y_2 \leq 1 \\ & y_1 \leq 0, \ y_2 \geq 0 \end{aligned}$$

Since $x_2^* < 0, x_3^* < 0$, the last two constraints should be tight, $y_1 + Ky_2 = -2, -y_1 - y_2 = 1$.
Then we know $-y_1 - y_2 = 1$, optimal value $4y_1 + 6y_2 = -3.5$, so that $y_1^* = -1.25$,$y_2^* = 0.25$, K=-3

Exercise 5

The optimal solution of LP dual problem is $y^* = (1.2, 0.2)^T$. And the LP primal problem is

$$\begin{aligned} \text{Max}_x \ & x_1 + 2x_2 + 3x_3 + 4x_4 \\ \text{subject to } & x_1 + 2x_2 + 2x_3 + 3x_4 \leq 20 \\ & 2x_1 + x_2 + 3x_3 + 2x_4 \leq 20 \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0,\ x_4 \geq 0 \end{aligned}$$

What is the optimal solution of primal problem x^*?

Exercise 5 Solution

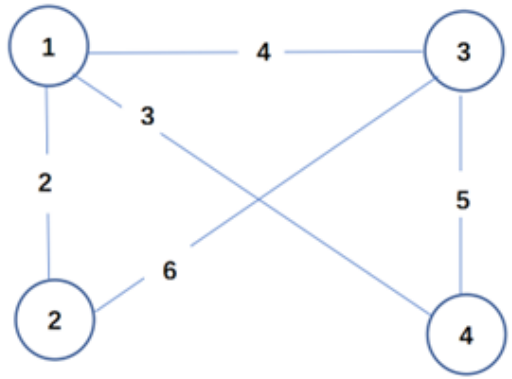
The LP dual formulation is

$$\begin{aligned} \text{Min}_y \ & 20y_1 + 20y_2 \\ \text{subject to } & y_1 + 2y_2 \geq 1 \\ & 2y_1 + y_2 \geq 2 \\ & 2y_1 + 3y_2 \geq 3 \\ & 3y_1 + 2y_2 \geq 4 \\ & y_1 \geq 0, \ y_2 \geq 0 \end{aligned}$$

Since $y_1^* = 1.2 > 0, y_2^* = 0.2 > 0$, two constraints of primal problem should be tight. $x_1^* + 2x_2^* + 2x_3^* + 3x_4^* = 20$, $2x_1^* + x_2^* + 3x_3^* + 2x_4^* = 20$
And $y_1 + 2y_2 = 1.6 > 1$, $2y_1 + y_2 = 2.6 > 2$, $2y_1 + 3y_2 = 3$, $3y_1 + 2y_2 = 4$.The first two constraints of dual problem should be relaxed, while the last two constraint of dual problem should be tight. So $x_1^* = x_2^* = 0$.
Then we get $2x_3^* + 3x_4^* = 20$, $3x_3^* + 2x_4^* = 20$, $x^* = (0, 0, 4, 4)$

Exercise 6

Maximum flow problem: Consider the maximum flow problem in the following network, the flow paths form 1 to 4 are [(1,2)(2,3)(3,4)],[(1,3)(3,4)],[(1,4)] respectively. Each edge has a capacity constraint.



If we assume the flow is x_1, x_2, x_3, formulate your primal problem and dual problem. What is the meaning of dual problem?

Exercise 6 Solution

The primal formulation is

$$\begin{aligned} \text{Max}_x \; & x_1 + x_2 + x_3 \\ \text{subject to } & x_1 \leq 2 \\ & x_2 \leq 4 \\ & x_3 \leq 3 \\ & x_1 \leq 6 \\ & x_1 + x_2 \leq 5 \\ & x_1 \geq 0, \; x_2 \geq 0, \; x_3 \geq 0 \end{aligned}$$

The primal optimal solution is $x^* = (1, 4, 3)$

Exercise 6 Solution

The dual formulation is

$$\begin{aligned} \text{Min}_y \ & 2y_1 + 4y_2 + 3y_3 + 6y_4 + 5y_5 \\ \text{subject to } & y_1 + y_4 + y_5 \geq 1 \\ & y_2 + y_5 \geq 1 \\ & y_3 \geq 1 \\ & y_1 \geq 0,\ y_2 \geq 0,\ y_3 \geq 0,\ y_4 \geq 0,\ y_5 \geq 0 \end{aligned}$$

The dual optimal solution is $y^* = (0, 0, 1, 0, 1)$. This is a minimum cut problem. We cut edge (1,4) and (3,4) to separate point 1 and 4.

Acknowledgement

Thank you for your time!

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